1) Only text in double quotes appears on the screen. The rest is for ChatGPT.
2) Text in single quotes is the name of the button. The text following it in parentheses and double quotes is the text that appears on the button.
3) A pound sign/number sign (#) serves as the “if” and the chevron sign (>) serves as “then.” The plus sign (+) denotes the next screen. For example, “# German > +Screen 0002+ means that if the user selects the *German* button, they’re taken to Screen 0002.
4) *Cards* are treated as types of *screens*, specifically screens that appear during drills or study mode. I’m doing it this way to keep the numbering simple and consistent. i.e. so I can say “screen 0001” and “card 3201” or something.
5) [Temporary] — display screen/card titles at top left of screen/card
6) Update the existing unique identifiers (ten digits flanked by a string of ampersands). Whenever a chunk of code pertains to a particular card, make the unique identifier for that chunk a ten-digit code starting with the card ID. For the first instance of a chunk pertaining to that card, make the next six digits in the code “XXXXXX” (i.e. 3201XXXXXX). If there are multiple chunks for a particular card, each subsequent chunk will replace the first X with a letter, i.e. “3201AXXXXX” for the first, “3201BXXXXX” for the second, and so on. If there are more than 27 chunks pertaining to a specific card, the format will be “3201AAXXXX,” “3201ABXXXX,” and so on.

CODE ARCHITECTURE NOTE — UNIQUE IDENTIFIERS

Use a hybrid identifier system.

1) SCREEN/CARD IDENTIFIERS
Each screen or card should have its own internal ID based on its UX flow name/number.
Example:
- SCREEN_0001_HOME
- SCREEN_0204_DRILL_SETUP
- CARD_2103_GENDER_RECALL

These identifiers should govern:
- screen/card rendering
- visible text for that screen/card
- local button wiring
- local transitions

2) SHARED SYSTEM IDENTIFIERS
Do not assume all logic for a screen/card lives in one place.
Some behavior is governed by shared systems used by many screens/cards.
These systems should have their own identifiers.

Example:
- SYS_NAVIGATION
- SYS_APP_STATE
- SYS_SAVE_LOAD
- SYS_SESSION_BUILDER
- SYS_CARD_GENERATION
- SYS_ANSWER_VALIDATION
- SYS_WEIGHTING
- SYS_REPLACEMENT_QUEUE
- SYS_GLOBAL_UI_RULES

These identifiers should govern:
- navigation rules
- persistence
- scoring / error count / cards remaining
- drill/session construction
- validation
- weighting / frequency rules
- replacement flows
- cross-screen UI rules

3) CONFIG / REGISTRY IDENTIFIERS
Global mappings and constants should also be separate.
Example:
- CFG_SCREEN_REGISTRY
- CFG_LANGUAGE_RULES
- CFG_COMPETENCIES
- CFG_CARD_TYPES
- CFG_TEXT_RULES
- CFG_COLOR_RULES

WORKING RULE
A screen/card ID should identify the place where that screen/card is rendered.
It should NOT be treated as the single home for every rule that affects that screen/card.

WHY
Some screen behavior is local, but some is controlled by global state variables, shared UI rules, and shared drill/study logic.
Therefore, unique identifiers should be based partly on screen/card names and partly on shared systems.

RECOMMENDATION
Use screen/card names for render chunks.
Use separate system identifiers for anything reused across multiple screens/cards.

This will make surgical editing easier and prevent confusion later.

SYSTEM STATE VARIABLES

languageSelected

studyTypeSelected

direction_mode

currentCard

errorCount

cardsRemaining

SCREEN TYPES

0000–0099 Global navigation screens
0100–0199 Study selection
1000–1999 Study cards
5000–5999 Management / List

CODE

= if
> = go to

SCREEN 0001 — HOME

TYPE

Navigation / Language Select

VISIBLE TEXT
"Welcome!"
"Please choose your language"

BUTTON GROUP: LANGUAGE_SELECTION

ID: Spanish
Label: "Spanish"
Flag: Spain

ID: French
Label: "French"
Flag: France

ID: German
Label: "German"
Flag: Germany

UI STYLE

Buttons: rectangular with rounded corners

GRAPHICS STYLE

Flag position: flush left, occupying the left side of the button
Flag height: same as button height
Flag aspect ratio: preserved
Flag alignment: flush with top and bottom edges of button
Flag corners: top-left and bottom-left clipped by button’s rounded corners

Label font: bold
Label size: medium
Label position: to the right of the flag

Button alignment: centered
Button spacing: medium

TRANSITIONS

Spanish > Screen 0101.0
French > Screen 0101.0
German > Screen 0101.0

STATE UPDATE ON CLICK

languageSelected = [Spanish | French | German]
studyTypeSelected = null

SCREEN 0005 — ADD CUSTOM WORD

TYPE
Input / Form

VISIBLE TEXT
"Add a word to your vocabulary"

LANGUAGE CONTEXT
Inherited from current deck

LAYOUT

Top:
- Title: "Add a new vocabulary word"

Middle:
- Form fields

Bottom:
- Button: "Save"
- Button: "Cancel"

FORM FIELDS

Text box:
- Target language word

Text box:
- English translation

Drop-down:
- Part of speech
Help icon:

- "Choose the word type. If you are unsure, check a dictionary."
Help icon:
- "Prepositions are not included here because they do not map cleanly between languages and work better in context than as standalone flashcards."

Drop-down:
- Gender (if applicable)

Text box:
- Plural form (if applicable)

Drop-down:
- Category (optional)

FIELD RULES

Target language word:
- User must enter the base form (lemma) only
- Do NOT include articles or determiners
- This rule applies to all languages

English translation:
- Required
- If multiple accepted English meanings apply, enter them in the same field separated by commas
- Each comma-separated value is treated as an individually accepted answer
- Users are not expected to type multiple meanings in a single response

English translation (verbs):

- May be entered as either:
  - bare form (e.g., "eat")
  - infinitive form (e.g., "to eat")

- The system will treat both forms as equivalent during answer validation
- Users will not be penalized for including or omitting "to"

Part of speech:
- Required

Gender:
- Required for nouns where applicable
- Hidden or disabled for parts of speech where gender does not apply

Plural form:
- Optional or required depending on language/POS rules

Category:
- Optional
- User may:
  - choose an existing category
  - create a new category
  - skip

Target language input must use correct spelling, including accents/diacritics.

- Do NOT enter anglicized forms (e.g., "ueber")
- Use the correct native spelling (e.g., "über")

If multiple English meanings are entered:

- Separate them with commas
- Each meaning will be accepted individually as a correct answer
- Users are not expected to type multiple meanings in a single response

IMPORTANT

- The canonical stored form is always the bare lemma (no articles or determiners)
- Articles and possessive pronouns are not stored in the noun lemma
- Card display forms may include definite or indefinite articles depending on card rules
- User-added words must preserve the same bare-lemma storage convention as the master dataset
- If article-bearing display forms are needed, the system must generate or store them separately from the canonical lemma
- Comma-separated English values are used only for answer validation
- They do NOT change the canonical stored lemma
- Category data is only useful if SCREEN 5001 later supports category-based vocab-bank control

LANGUAGE-SPECIFIC LEMMA RULES

German:
- Nouns must be entered as bare lemmas only
- Do NOT include der / die / das
- Example valid inputs:
  - Mann
  - Frau
  - Kind

French:
- Nouns must be entered as bare lemmas only
- Do NOT include le / la / l’
- Example valid inputs:
  - homme
  - femme
  - garçon

Spanish:
- Nouns must be entered as bare lemmas only
- Do NOT include el / la
- Example valid inputs:
  - hombre
  - mujer
  - niño

VERBS
- Enter infinitive / lemma form only

ADJECTIVES
- Enter base lemma form only

DISPLAY-FORM STORAGE RULE

For nouns, the system must preserve:

1. CANONICAL LEMMA
- bare noun only
- examples:
  - German: Hund
  - French: chien
  - Spanish: perro

2. DISPLAY FORMS
These are generated or stored separately for cards that require articles.

French examples:
- definite singular display form: le chien / la table
- indefinite singular display form: un chien / une table

Spanish examples:
- definite singular display form: el perro / la mesa
- indefinite singular display form: un perro / una mesa

German examples:
- article-bearing display forms may be generated separately where needed
- canonical lemma remains bare noun only

RULE
- Card-generation logic may use display forms
- Canonical storage must always preserve the bare lemma

BEHAVIOR / LOGIC

SAVE VALIDATION RULES

System must reject:
- incomplete entries
- malformed entries
- invalid part-of-speech / gender combinations
- noun entries missing required gender metadata
- duplicate entries if duplicate handling rules forbid them

System must normalize comma-separated English values:

english:
- split by commas for validation purposes
- trim whitespace
- remove empty entries
- remove duplicates
- preserve the full comma-separated string in the english field unless implementation stores a normalized internal answer list separately

LANGUAGE-SCHEMA VALIDATION

German nouns:
- Reject entries that include articles
- Require gender selection

French nouns:
- Reject entries that include definite articles
- Require gender selection

Spanish nouns:
- Reject entries that include definite articles
- Require gender selection

Verbs:
- Require lemma / infinitive form only

Adjectives:
- Require lemma form only

SYSTEM INTEGRATION

On successful save:
- add word to MASTER DATASET
- mark word as user_added = "y"

If launched from vocab bank management flow:
- allow immediate insertion into ACTIVE VOCAB BANK

This screen must follow:
- SYSTEM RULES — ISOLINGO (V0)

User-added entries must:
- match the schema of the language's existing master dataset
- preserve bare-lemma storage conventions
- remain compatible with SCREEN 5001 — VOCAB BANK MANAGER

SCHEMA COMPATIBILITY PATCH

PURPOSE
Ensure SCREEN 0005 produces records that match the canonical V0 vocabulary schema.

USER-ENTERED FIELDS FROM SCREEN 0005
- lemma
- english
- partOfSpeech
- gender (if applicable)
- plural (if applicable)
- category (optional)

SYSTEM-GENERATED FIELDS
- id
- targetLanguage
- competencies

SCHEMA MAPPING RULES

lemma

→ map directly from "Target language word"

english
→ map directly from "English translation"
→ if multiple meanings are entered, they remain as comma-separated values
→ validation logic handles splitting and matching

partOfSpeech
→ map directly from POS dropdown

gender
→ map from Gender dropdown when applicable
→ otherwise null

plural
→ map from Plural field when applicable
→ otherwise null

category
→ map from Category dropdown if selected
→ otherwise default to a general category

competencies
→ system-generated from partOfSpeech
- noun → ["vocab","gender"]
- non-noun → ["vocab"]

targetLanguage
→ system-generated from current language context
→ allowed values:
 - "de"
 - "fr"
 - "es"

id
→ system-generated

senseLabel
→ default null
→ used only when needed to distinguish true multiple meanings of the same lemma

RULE
SCREEN 0005 must only produce fields defined in the canonical V0 schema.
No additional accepted-answer fields are stored at the schema level.

INTERACTION

Save behavior:
- New custom word is added to the MASTER DATASET
- New custom word is flagged:
  - user_added = "y"
- New custom word becomes eligible for ACTIVE VOCAB BANK insertion
- If this screen was opened from SCREEN 5001 — VOCAB BANK MANAGER:
  - the new custom word is inserted into the ACTIVE VOCAB BANK immediately
  - one existing word is removed using replacement rules if needed to preserve bank size

Cancel behavior:
- Return without saving
- No changes are made to MASTER DATASET
- No changes are made to ACTIVE VOCAB BANK

TRANSITIONS

Save > Screen 5001
Cancel > Screen 5001

SCREEN 0101.0 — STUDY TYPE SELECT

TYPE
Navigation / Study Type Select

LANGUAGE CONTEXT
Inherited from Home

VISIBLE TEXT
"[Selected language]"
"Choose what you want to study"

BUTTON GROUP: PRIMARY_ACTIONS

ID: General_Vocab
Label: "General Vocab"

ID: Gender
Label: "Gender"

DIRECTION SELECTION

Radio button group: DIRECTION_MODE

Options:

( ) Target → Native only
( ) Native → Target only
( ) Mixed (random)

DEFAULT:
- Mixed (random) is selected by default

SYSTEM STATE VARIABLES

languageSelected
studyTypeSelected
direction_mode
currentCard
errorCount
cardsRemaining

BUTTON GROUP: SECONDARY_ACTIONS

ID: Home
Label: "Home"

UI STYLE

PRIMARY_ACTIONS
Shape: rectangular with rounded corners
Label font: bold
Label size: medium
Position: centered, stacked
Size: large
Spacing: medium

SECONDARY_ACTIONS
Shape: rectangular
Label font: regular
Label size: small
Position: bottom of screen
Size: small
Spacing: medium

GRAPHICS STYLE
Flag position: left of language text

TRANSITIONS

General_Vocab > Screen 0104
Gender > Screen 0104
Home > Screen 0001

On continue:
- pass direction_mode to study session engine

STATE UPDATE ON CLICK

General_Vocab
studyTypeSelected = "general_vocab"

Gender
studyTypeSelected = "gender"

Target → Native only
direction_mode = "target_to_native"

Native → Target only
direction_mode = "native_to_target"

Mixed (random)
direction_mode = "mixed"

SCREEN 0104 — SYSTEM — STUDY SESSION ENGINE

TYPE
System / Runtime

PURPOSE
Generate and deliver Study cards based on:
- Study Repetition Engine
- Study Card Rotation System
- Selected study type
- Active Vocab Bank

BEHAVIOR

On entry:
→ Initialize study session

→ Apply study-type filter

IF studyTypeSelected = "general_vocab":
  - eligible words = nouns, verbs, adjectives, adverbs
  - nouns remain color-coded where applicable
  - card pool = general vocab cards only

IF studyTypeSelected = "gender":
  - eligible words = nouns only
  - card pool = gender cards only

→ Select first card dynamically using:
  - pool distribution (active / reinforcement / variation)
  - repetition engine weighting
  - card rotation rules
  - anti-autopilot constraints

SYSTEM RESPONSIBILITY SPLIT

- Repetition Engine:
  determines WHICH vocabulary or pattern appears

- Card Rotation System:
  determines WHICH CARD TYPE is used to present that content

- Active Vocab Bank:
  provides available lexical items

DIRECTION MODE LOGIC

direction_mode controls prompt/answer direction for all eligible one-way translation cards.
Some cards may be direction-neutral and handled by card-specific compatibility rules.

target_to_native:
- All eligible one-way translation cards use Target → Native format
- Direction-neutral cards are excluded

native_to_target:
- All eligible one-way translation cards use Native → Target format
- Direction-neutral cards are excluded

mixed:
- System randomly selects direction per card
- Must avoid long streaks of the same direction
- Recommended:
  - no more than 3 consecutive cards in the same direction

TARGET_TO_NATIVE LOW-VALUE CARD EXCLUSION

When direction_mode = target_to_native:

The system must exclude any card whose target-language prompt is identical or near-identical to the English answer in a way that makes the answer obvious without meaningful retrieval.

Examples:
- Hand → hand
- Jeans → jeans
- Jobinterview → job interview

RULES:
- Exclude these only in target_to_native mode
- Allow them in native_to_target mode
- If a selected item is excluded, replace it with another eligible item
- This rule applies before card rendering

CARD-SPECIFIC DIRECTION COMPATIBILITY

CARD 1001.0 — Matching (Gender Anchoring)
- is direction-neutral, not one-way translation
- tests noun–gender matching / recognition

CARD 1002 — Vocab Relationship Judgment
- is direction-neutral, not one-way translation
- compares an English word and a target-language word on the same card

RULES:
- In mixed mode:
  - CARD 1001.0 remains eligible
  - CARD 1002 remains eligible
- In target_to_native mode:

  - CARD 1001.0 is NOT eligible
  - CARD 1002 is NOT eligible
- In native_to_target mode:
  - CARD 1001.0 is NOT eligible
  - CARD 1002 is NOT eligible

RATIONALE:
- Pure direction modes should contain only cards with a clear one-way prompt/answer direction
- CARD 1001.0 and CARD 1002 do not test one-way translation

STUDY CARD POOLS (LANGUAGE-SPECIFIC)

The Card Rotation System may only select from the card pool allowed by the currently selected study competency.

GERMAN — VOCAB & GENDER
Eligible Study cards:
- CARD 1001 — Matching (Gender Anchoring)
- CARD 1002 — Vocab Relationship Judgment
- CARD 1006 — Noun Production (Language-Specific)

SPANISH / FRENCH — VOCAB & GENDER
Eligible Study cards:
- CARD 1001 — Matching (Gender Anchoring)
- CARD 1002 — Vocab Relationship Judgment
- CARD 1006 — Noun / Determiner Production

RULES:
- use the French / Spanish behavior defined inside each card’s language-specific rules
- do not use German-only case or adjective-ending logic

Each card must support both directions where applicable.

If a card type does not support the selected direction:
- system must skip or replace with a compatible card

RULES

- Only cards within the active competency set are eligible
- The system must NOT pull cards from a different competency pool
- If multiple competencies are active in Study Mode, the Card Rotation System may alternate between the eligible pools

- Card selection must still respect:
  - repetition engine weighting
  - anti-autopilot constraints
  - variation rules

GENDERED NOUN PROMPT RULE

For nouns with distinct masculine and feminine forms referring to people:

TARGET → NATIVE:
- Select and display a single noun form
- Do NOT combine multiple forms in one prompt
- Accept natural English equivalents
- Accept gender-explicit English equivalents when correct for the displayed target-language form

Examples:
- Ärztin → "doctor" OR "female doctor"
- Kellnerin → "waitress" OR "server" OR "female server"
- ciudadana → "citizen" OR "female citizen"

NATIVE → TARGET:
- Use a disambiguating English prompt when needed to ensure a single correct answer
- Use natural English gendered terms where available
- Use gender-explicit prompts when English does not naturally distinguish the forms

Examples:
- "waiter" → Kellner
- "waitress" → Kellnerin
- "female doctor" → Ärztin
- "male citizen" → ciudadano
- "female citizen" → ciudadana

RULES:
- Avoid ambiguous English prompts unless multiple target answers are explicitly allowed
- The prompt must guide the user toward the intended target-language form

SAME-LEMMA ANSWER POOL RULE

For target_to_native cards:

Before rendering a target-language prompt, the system must check whether multiple active vocab entries share the same:
- targetLanguage
- normalized lemma

If yes:
- build a shared English answer pool from all matching entries
- validate the user's answer against the full shared pool

FEEDBACK RULE:
- If the user enters any valid English answer from the shared pool:
  - mark the answer correct
  - display:
    "Also correct: [other accepted meanings]"

COMMA-SEPARATED ENGLISH VALUES RULE

If a vocab entry contains multiple comma-separated English values:

- Treat each value as an independent valid answer
- Do NOT require the user to enter multiple values

VALIDATION:
- Accept any single value as correct

FEEDBACK:
- After a correct answer, display:
  "Also correct: [other values from the same entry]"

EXAMPLE:
english: "free, available"

User enters:
- "free" → Correct
  "Also correct: available"

RULES:
- Do NOT require the user to guess which English meaning the system intended
- Do NOT mark a same-lemma English meaning incorrect
- The current card's primary English gloss remains the main answer
- Other meanings appear only in the "Also correct" line

LOOP

After each card:
→ Update:
  - exposureCount
  - errorCount

- frequencyWeight

→ Select next card dynamically using:
- repetition engine (which item/pattern)
- active competency pool (which set of card types is eligible)
- card type rotation rules (which card type within that pool)
- variation constraints
- anti-autopilot constraints (no card type repetition patterns)

→ Load next card

EXIT

On Exit Study:
→ SCREEN 0001 — HOME

CARD 1001.0 — STUDY MODE MATCHING (GENDER ANCHORING)

TYPE
Study Card — Matching

MODE
Recognition (Target Language Focus)

DIRECTION MODE COMPATIBILITY

This card is direction-neutral.

It does NOT operate as:
- Target → Native only
- Native → Target only

It operates as:
- target-language gender matching / recognition

RULES:
- Eligible in mixed mode
- Not eligible in pure target_to_native mode
- Not eligible in pure native_to_target mode

MODE CLASSIFICATION
Study Mode only

COMPETENCY
Gender (Primary)
Vocab (Secondary)

PURPOSE
Reinforce noun–gender associations through fast, repeated exposure using article support and color anchoring.

Build automatic recognition of noun gender without requiring production.

LANGUAGE CONTEXT
Spanish / French / German

VISIBLE TEXT

Top Left:
[Language flag icon]

Top Right:
Home button

Top Left (screen edge):
Exit Study button

Instruction Text (centered):
"match the words"
- Font: normal
- Size: small-to-medium
- Tone: neutral

LAYOUT

Two-column matching interface:

LEFT COLUMN (PROMPTS)
- Target-language nouns
- Display rules depend on selected language:

German:
- Show bare noun only
- No article displayed
- Noun remains capitalized
- Noun is color-coded by gender

French:
- Show definite article + noun
- Example:
  - le chien
  - la femme
- Color applies only to the noun itself, not to the article

Spanish:
- Show definite article + noun
- Example:
  - el perro
  - la mujer
- Color applies only to the noun itself, not to the article

RIGHT COLUMN (ANSWERS)
- Color-coded gender tiles (no text)

German:
- Tile 1: Masculine (color #4da3ff)
- Tile 2: Feminine (color #d66aa6)
- Tile 3: Neuter (color #A4922D)

French / Spanish:
- Tile 1: Masculine (color #4da3ff)
- Tile 2: Feminine (color #d66aa6)

INTERACTION

- User taps one item from LEFT column
- Then taps one item from RIGHT column

→ A line or highlight connects the pair

- Matched pairs remain visible
- User continues until all items are matched

BUTTONS

Primary:
Submit (appears only after all pairs are matched)

Secondary:
Flip (always visible, below matching area)
Exit Study (top left)
Home (top right)

STATE → VISUAL OUTPUT

STATE: Matching
- Selections highlight as user taps
- Connections remain visible
- Submit disabled until all pairs assigned

STATE: First Submission — Correct
- Show:
  "Correct!"
- Submit replaced with green inactive button
- "Next" button appears

STATE: First Submission — Incorrect
- Show:
  "Not quite — try again"
- Incorrect pairs visually reset
- Correct pairs remain locked
- User continues matching

STATE: Flip Activated
- All correct matches displayed immediately
- Incorrect matches corrected
- Input disabled
- Show "Next" button
- Submit removed

BEHAVIOR / LOGIC

CONTENT GENERATION

Step 1:
Select 3–5 nouns from active vocabulary bank

Step 2:
Ensure gender diversity

German:
- At least 2 different genders per card
- Prefer all 3 when possible

French / Spanish:
- Must include both masculine and feminine when possible

Step 3:
Generate display form for each noun based on language rules

German:
- bare noun only

French:
- definite article + noun display form

Spanish:
- definite article + noun display form

Step 4:
Shuffle noun order

Step 5:
Generate matching set

German:
- Masculine tile (#4da3ff)
- Feminine tile (#d66aa6)
- Neuter tile (#A4922D)

French / Spanish:
- Masculine tile (#4da3ff)
- Feminine tile (#d66aa6)

ANSWER VALIDATION

Correct if:
- All noun–gender pairs match gender metadata

Partial correctness:
- Preserve correct matches
- Reset incorrect ones

REPETITION INTEGRATION

- Incorrect nouns return to ACTIVE POOL
- Correct nouns move toward REINFORCEMENT POOL
- Same nouns reappear with variation (new groupings)

(See SYSTEM — STUDY MODE REPETITION ENGINE)

FLIP SYSTEM INTEGRATION

- Flip reveals all correct matches immediately
- No penalty
- Advances to next card

(See SYSTEM — STUDY MODE FLIP BEHAVIOR)

ANTI-AUTOPILOT RULES

- Do not repeat same noun group in short window
- Rotate noun categories
- Shuffle answer tile positions each time

TRANSITIONS

On Next:

→ Load next study card

On Exit:
→ SCREEN 0001 — HOME

NOTES FOR IMPLEMENTATION

- Each noun must include gender metadata
- German nouns must use the 3-gender system
- French and Spanish nouns must use the 2-gender system
- Color mapping must match global color system
- Matching state must track:
  - selected pairs
  - locked correct pairs
- Flip overrides all interaction states
- Submit only enabled when all pairs assigned

DESIGN GOAL

Maximize:
- rapid exposure
- gender recognition
- article + color association where applicable

Minimize:
- cognitive load
- friction
- production pressure

This is a pure encoding card, not a testing card.

CARD 1002 — STUDY MODE VOCAB RELATIONSHIP JUDGMENT

TYPE
Study Card — Multiple Choice (Relationship Judgment)

MODE

Study Mode only

DIRECTION MODE COMPATIBILITY

This card is direction-neutral.

It does NOT operate as:
- Target → Native only
- Native → Target only

It operates as:
- bilingual comparison / relationship judgment

RULES:
- Eligible in mixed mode
- Not eligible in pure target_to_native mode
- Not eligible in pure native_to_target mode

COMPETENCY
Vocabulary

LANGUAGE CONTEXT
Spanish / French / German

PURPOSE
Strengthen word association and discrimination by prompting the user to judge the relationship between an English word and a target-language word.

CORE DESIGN RULE
This card uses ONE fixed card type with controlled pair generation behind the scenes.
The card itself does not change type depending on the relationship being tested.

VISIBLE TEXT

Top Left:
[Language flag icon]

Top Right:
Home button

Top Left (screen edge):
Exit Study button

Instruction Text (centered, upper portion):

"what is the relationship between these words?"
- Font: normal weight
- Size: small-to-medium
- Tone: neutral, non-imposing

Word Pair Display (centered):
Line 1:
"[English word]"
- Font: medium-large
- Weight: bold

Line 2:
"[target language word]"
- Font: medium-large
- Weight: bold

IMPORTANT DISPLAY RULES

German:
- If the target-language word is a German noun, it remains capitalized
- German nouns are displayed without article on this card
- If the target-language word is a German noun, the noun itself is color-coded by default
- Color applies only to the noun itself, not to any surrounding text

French / Spanish:
- If the target-language word is a French or Spanish noun, it is displayed with an article
- The displayed article may be definite or indefinite
- The noun itself is color-coded by gender
- The article is not color-coded

Global:
- This card is about lexical relationship, not grammar retrieval
- It still participates in Study Mode gender exposure when nouns appear

ANSWER FORMAT

Multiple choice
Exactly four answer buttons are shown:

1) "They mean the same thing"
2) "They belong to the same category"
3) "They look or sound similar"
4) "No clear relationship"

BUTTON STYLE
- Rectangular with rounded corners
- Standard study-mode button styling
- Stacked vertically or displayed in a clean 2x2 layout
- Large enough for easy mobile tapping

SECONDARY ACTIONS
Flip
Home
Exit Study

POST-ANSWER BEHAVIOR

Correct Answer:
- Selected answer becomes inactive
- Feedback text appears:
  "Correct"
- Next button appears

Incorrect Answer:
- Selected answer becomes inactive
- Feedback text appears:
  "Not quite"
- The correct answer is visually indicated
- Next button appears

FLIP BEHAVIOR

If user taps "Flip":
- The correct answer is revealed immediately
- All answer buttons become inactive
- Feedback text appears:
  "Answer shown"
- Next button appears

NEXT BUTTON
Label:
"Next"

Behavior:
- Advances to next study card

BEHAVIOR / LOGIC

PAIR GENERATION MODEL

This card is driven by pair logic, not by separate card types.

For each card, the system generates one English word and one target-language word, then assigns exactly one intended relationship type.

ALLOWED RELATIONSHIP TYPES

1) SAME MEANING
Definition:
- The two words are translations of each other

Example:
- dog / Hund

Correct answer:
- "They mean the same thing"

2) SAME CATEGORY
Definition:
- The two words are not translations of each other, but they belong to the same broad semantic group

Example:
- dog / Katze
- bread / Milch

Correct answer:
- "They belong to the same category"

3) SIMILAR FORM
Definition:
- The two words are not translations of each other, but they visibly or audibly resemble one another enough to support noticing

Examples may include:
- cognates
- near-cognates
- words with obvious shared roots
- other high-confidence form similarities approved in the active vocabulary bank

Correct answer:
- "They look or sound similar"

IMPORTANT:
- This label appears on the card exactly as written
- No technical linguistic terminology appears on screen

4) NO CLEAR RELATIONSHIP
Definition:
- The two words are not translations
- They do not belong to the same category
- They do not have a meaningful form similarity

Example:
- cat / Vogel

Correct answer:
- "No clear relationship"

SEMANTIC CLARITY ENFORCEMENT

- Exactly one answer choice must be clearly correct
- Avoid ambiguous pairings where multiple answers could reasonably apply
- Synonym pairs may be used only when the correct answer is "They mean the same thing"

DATA / TAGGING REQUIREMENTS

Minimum metadata required for scalable implementation:

For each vocab item:
- English gloss
- Target-language form
- Category tag

Optional but strongly recommended for this card:
- Similar-form tag
- Similar-form group ID or pairing map

RECOMMENDED IMPLEMENTATION MODEL

Each generated card should begin by selecting an intended relationship type.

Then:

IF relationship type = SAME MEANING
- Select one vocab item

- Show its English word and its target-language translation

IF relationship type = SAME CATEGORY
- Select two different vocab items
- Ensure:
  - same category
  - not translations of each other
  - not also classified as similar-form pair

IF relationship type = SIMILAR FORM
- Select from a preapproved pool of similar-form pairs
- Do not rely on on-the-fly string comparison alone
- Pair must be intentionally flagged in active vocabulary bank or pairing logic

IF relationship type = NO CLEAR RELATIONSHIP
- Select two words that:
  - are not translations
  - are not in the same category
  - are not marked as similar-form pair

DISAMBIGUATION RULE
Each pair must resolve to one clearly best answer.
Do not generate pairs that could plausibly fit multiple answer choices unless one relationship type has been explicitly prioritized in the underlying logic.

PRIORITY RULE FOR OVERLAP
If a pair could fit more than one relationship type, use this priority order:

1) They mean the same thing
2) They look or sound similar
3) They belong to the same category
4) No clear relationship

This rule is internal only and not shown to the user.

CONTENT CONSTRAINTS

- Avoid rare, confusing, or highly debatable pairings in V0
- Avoid pairs that require specialist etymological knowledge
- Avoid weak or subjective "similar form" matches
- Prefer high-confidence, easy-to-justify examples
- Keep category relationships broad and intuitive (animals, food, places, people, etc.)

FEEDBACK RULES

Correct:
- "Correct"

Incorrect:
- "Not quite"

OPTIONAL ENHANCEMENT FOR LATER VERSION
After answer submission, a short explanation may appear under feedback, such as:
- "These are translations."
- "These are both animals."
- "These look similar."
- "These are unrelated."

This explanation layer is optional and not required for V0.

STATE RULES

Track per card:
- relationship type
- correct answer
- selected answer
- correct / incorrect result

Internal tagging (not visible to user):
- same_meaning
- same_category
- similar_form
- no_clear_relationship

TRANSITIONS

On Next:
→ Load next study card

On Exit:
→ SCREEN 0001 — HOME

On Home:
→ SCREEN 0001 — HOME

NOTES FOR IMPLEMENTATION

- This is a Study Mode card, not a Drill card

- The goal is noticing and discrimination, not forced production
- This card should be mixed into study sessions as one vocabulary-support card type among others
- Best used at moderate frequency, not as the dominant vocab card
- Safest launch set:
  - same meaning
  - same category
  - no clear relationship
- Similar-form pairs should be included only when they are manually approved or explicitly tagged

CARD 1006 — STUDY MODE NOUN PRODUCTION (LANGUAGE-SPECIFIC)

TYPE
Study Card — Translation + Gender Binding

LANGUAGE CONTEXT
German / French / Spanish

MODE
Direction-sensitive

Native → Target:
- English → Target Language

Target → Native:
- Target Language → English

MODE CLASSIFICATION
Study Mode only

DIRECTION MODE COMPATIBILITY

This card is a one-way translation card whose prompt and expected answer change based on direction_mode.

RULES:
- Eligible in target_to_native mode
- Eligible in native_to_target mode
- Eligible in mixed mode

COMPETENCY
Vocab + Gender

INTERACTION
Uses: SYSTEM RULES — ISOLINGO (V0), Section 11 — NOUN RESPONSE RULE
Uses: SYSTEM RULES — ISOLINGO (V0), Section 14 — FLIP BEHAVIOR

PURPOSE
Train direct recall of target-language noun production using the language-appropriate gender mechanism.

Reinforce:

German:
- English → German noun retrieval
- noun–gender binding
- reflexive association between noun form and gender color

French / Spanish:
- English → target-language determiner + noun retrieval
- determiner-based gender encoding
- reflexive association between noun form and determiner choice

---

VISIBLE TEXT

Top Left:
[Language flag icon]

...

PROMPT / RESPONSE MATRIX

This card is direction-sensitive.

Direction and language determine:
- the prompt format
- the expected input format
- whether gender tiles appear
- how validation works

NATIVE → TARGET

German:
Prompt:
- English noun only
Examples:
- "dog"
- "woman"
- "child"

Expected input:
- bare German noun only

Gender behavior:
- gender tile REQUIRED
- noun + gender tile are validated together

French / Spanish:
Prompt:
- English determiner + noun
Examples:
- "the dog"
- "a cat"
- "my friend"

Expected input:
- determiner + noun

Gender behavior:
- NO gender tile
- gender is encoded in determiner choice

TARGET → NATIVE

German:
Prompt:
- bare German noun only
Examples:
- "Hund"

- "Frau"
- "Kind"

Expected input:
- English translation

Gender behavior:
- NO gender tile
- card functions as translation recall only

French / Spanish:
Prompt:
- determiner + noun
Examples:
- "le chien"
- "la femme"
- "el perro"

Expected input:
- English translation

Gender behavior:
- NO gender tile
- card functions as translation recall only

Native → Target examples:
German:
- "dog"
- "woman"
- "child"

French / Spanish:
- "the dog"
- "a cat"
- "my friend"

Target → Native examples:
German:
- "Hund"
- "Frau"
- "Kind"

French / Spanish:
- "le chien"
- "la femme"
- "el perro"

- Font: slightly larger than instructions
- Position: centered

IMPORTANT DISPLAY RULE

German:
- No target-language text is shown before submission
- No color coding is shown before the user selects a gender tile
- The noun the user types remains inside the input field
- After the user selects a gender tile, the typed noun itself changes to that color
- The system must NOT render a second copy of the noun elsewhere on the card

French / Spanish:
- No target-language answer is shown before submission
- No gender tiles are shown
- The phrase the user types remains inside the input field
- The system must NOT render a second copy of the typed response elsewhere on the card
- If color appears in answer display or feedback, only the noun is color-coded, not the determiner

---

INPUT

- Single text input field

Placeholder text:

German:
"Type the German noun"

French / Spanish:
"Type the full phrase"

Expected:

German:
- bare noun only
- no article

- noun must remain capitalized

French / Spanish:
- determiner + noun
- determiner must match prompt type

DIRECTION VARIATION

Native → Target:
- User produces target-language answer

Target → Native:
- User produces English translation

RULE:
- Card behavior must switch input expectations based on direction_mode

---

LANGUAGE VARIATION

Native → Target:

German:
- Prompt: English noun only
- Input: bare German noun
- Gender tile required

French / Spanish:
- Prompt: English determiner + noun
- Input: determiner + noun
- No gender tile shown

Target → Native:

German:
- Prompt: bare German noun only
- Input: English translation
- No gender tile shown

French / Spanish:
- Prompt: determiner + noun

- Input: English translation
- No gender tile shown

German:
- Prompt: English noun only (e.g. "dog")
- Input: bare noun only
- Gender tile selection REQUIRED
- Answer = noun + gender tile (combined validation)

Spanish / French:
- Prompt: English determiner + noun
  - Examples:
    - "the dog"
    - "a cat"
    - "my friend"

- Input: determiner + noun (full phrase)
  - Examples:
    - el perro / la mesa
    - un chat / une femme
    - mon ami / ma sœur

- Gender tile selection NOT shown

- Answer = determiner + noun (single string validation)

- Gender is assessed implicitly through:
  - correct determiner selection

GENDER TILE SELECTION

German only:

Shown below input field:

- Tile 1: Masculine (#4da3ff)
- Tile 2: Feminine (#d66aa6)
- Tile 3: Neuter (#A4922D)

RULE:
- One tile must be selected
- Submit remains disabled until:

  - input is non-empty
  - one tile is selected

French / Spanish:
- No gender tiles are shown
- Submit becomes available when the determiner + noun input is non-empty

---

BUTTONS

Primary:
Submit

Secondary:
Flip (always visible)
Exit Study
Home

STATE → BEHAVIOR

STATE: Typing

German:

- Input updates live

- No feedback shown

- Tile selection available

- Submit disabled until:

  - input is non-empty

  - one tile is selected

French / Spanish:

- Input updates live

- No feedback shown

- No gender tiles are shown

- Submit disabled until:

  - input is non-empty

STATE: Tile Selected

German only:

- Selected tile is highlighted

- Typed noun changes to selected tile color

- Input field remains visible

- No duplicate noun rendering appears

French / Spanish:

- This state does not apply

STATE: Submission — Correct
- Show:
  "Correct!"

- Replace Submit with inactive "Correct!"
- Show "Next" button
- Typed noun remains color-coded in the selected correct gender color

---

STATE: Submission — Incorrect

German:

- Show:

  "Not quite — try again"

- Input remains active

- Tile selection remains active

- User may correct:

  - the noun

  - the gender tile

  - or both

French / Spanish:

- Show:

  "Not quite — try again"

- Input remains active

- No tile selection is shown

- User may correct:

  - the determiner

  - the noun

  - or both

STATE: Second Attempt — Incorrect
- Show:
  "Correct answer: ____"

- Display the correct German noun once
- Correct noun is color-coded by gender
- Input becomes inactive
- Show "Next" button

STATE: Flip Activated

German:

- Show:

  "Answer shown"

- Display correct German noun once

- Correct noun is color-coded by gender

- Input becomes inactive

- Tile selection becomes inactive

- Show "Next" button

French / Spanish:

- Show:

  "Answer shown"

- Display the correct determiner + noun once

- Noun is color-coded by gender

- Article/determiner is not color-coded

- Input becomes inactive

- Show "Next" button

BEHAVIOR RULES

- Max 2 attempts
- No penalties
- Flip available at all times
- Study Mode feedback remains soft

DETERMINER MATCH RULE

For French / Spanish:

The determiner used in the answer must match the determiner type in the prompt.

Examples:

Prompt: "the dog"
→ requires definite article

Prompt: "a dog"
→ requires indefinite article

Prompt: "my dog"
→ requires possessive

Incorrect:
- correct noun with wrong determiner type

LANGUAGE-SPECIFIC VALIDATION

German:
Correct if:
1) typed noun is correct
2) selected gender tile is correct

Spanish / French:
Correct if:
1) noun is correct
2) determiner agrees with noun gender

Incorrect if:
- wrong noun
- correct noun with incorrect determiner
- missing determiner

TARGET → NATIVE VALIDATION

Correct if:
- the English answer matches the canonical English translation
- or an approved additional accepted English meaning

Not required:
- gender tile
- determiner selection
- target-language article production

- For verbs, accept both bare verb form ("eat") and infinitive form ("to eat") as equivalent

Same-lemma answer handling:
- If the user enters an English answer associated with another active vocab entry that shares the same targetLanguage + lemma, mark correct
- Show:
  "Correct"
  "Also correct: [other accepted English meanings]"

RULE:
- In target_to_native mode, noun gender is not separately tested
- The card functions as translation recall only

GERMAN ANSWER VALIDATION

Correct if BOTH are correct:
1. typed noun matches the correct German noun
2. selected gender tile matches the noun's gender

Accepted lexical behavior:
- allow accepted lexical variants if defined in validation system

Not accepted:
- article + noun
- lowercase German noun
- correct noun with wrong gender tile
- wrong noun with correct gender tile

GERMAN CONTENT GENERATION RULES

STEP 1:
Select noun from ACTIVE VOCAB BANK

STEP 2:
Generate English prompt from noun's English field

STEP 3:
Require bare German noun production only

STEP 4:
Require matching gender tile selection

SPANISH / FRENCH PROMPT GENERATION

Step 1:
Select noun from ACTIVE VOCAB BANK

Step 2:
Select determiner type:
- definite article (the)
- indefinite article (a/an)
- possessive (my)

Step 3:
Generate English prompt:
- "the [noun]"
- "a [noun]"
- "my [noun]"

Step 4:
Require corresponding target-language output:
- Spanish:
  - el / la / un / una / mi
- French:
  - le / la / un / une / mon / ma

CONSTRAINTS

German:
- Noun only
- No article
- No possessive pronoun

French / Spanish:
- Determiner + noun required
- Determiner must match prompt type
- No noun-only responses allowed

Target → Native mode:

- No gender enforcement required
- No determiner enforcement required
- Accept valid English equivalents

TARGET_TO_NATIVE CONSTRAINT

In target_to_native mode, do not generate cards where the target-language prompt is identical or near-identical to the English answer in a way that makes the answer obvious.

These items remain allowed in native_to_target mode.

REPETITION ENGINE INTEGRATION

- Incorrect items return to ACTIVE POOL
- Correct items move toward REINFORCEMENT POOL

(See SYSTEM — STUDY MODE REPETITION ENGINE)

TRANSITIONS

On Next:
→ Load next study card

On Exit Study:
→ Return to Study menu

On Home:
→ Return to home screen

NOTES FOR IMPLEMENTATION

German:
- Must use German noun capitalization rules
- Must use noun gender metadata
- Input field text must change color directly after tile selection
- No duplicate noun rendering anywhere on the card

French / Spanish:

- Must use noun gender metadata
- Must require determiner + noun output in native_to_target mode
- Must not show gender tiles
- Must not color the determiner
- Must color only the noun where color display applies

This is the primary Study Mode noun-production card for vocab/gender across supported languages.

DESIGN GOAL

Maximize:
- noun recall
- noun–gender binding
- clean visual reinforcement
- low-friction productive repetition

Minimize:
- redundant UI
- duplicate rendering
- article interference
- unnecessary complexity

MENTAL MODEL

"I know the German noun, and I know its gender as part of the same answer."

SCREEN 5001 — VOCAB BANK MANAGER

SCREEN 5001 — VOCAB BANK MANAGER

TYPE
Management / List

LANGUAGE CONTEXT
Inherited from current language context

PURPOSE

Allow the user to:
- view current active vocabulary bank

- remove words
- replace words
- add new words
- understand what they are currently studying

This screen is the user-facing interface for:

SYSTEM — ACTIVE VOCAB BANK (V1)

VISIBLE TEXT

Header (centered):
"Your Vocabulary"

Subtext (centered):
"These are the words you are currently studying"

Optional helper text (small):
"Edit your list to control what appears in your study session"

LAYOUT

TOP BAR

Left:
Back button

Right:
Home button

MAIN AREA — WORD LIST

Scrollable vertical list

Each row displays ONE vocab item

WORD ROW STRUCTURE

LEFT SIDE

Primary text (bold):
[target-language word]

Secondary text (below):
[English translation]

Optional metadata (small, inline or below):
- part of speech
- gender (if noun, via color or label)

RIGHT SIDE — ACTIONS

Button: "Remove"

Style:
- small
- secondary (not dominant)

VISUAL STATES

Default:
- clean list
- evenly spaced rows

Optional (future):
- color indicator for gender (German only)

BOTTOM ACTION BAR

Primary Button:
"Add Word"

Secondary Button:
"Done"

INTERACTION

TAP "Remove"

→ Row enters Remove Selected state
→ User must tap "Confirm" to complete removal
→ User may tap "Cancel" to return to normal state

(See SYSTEM — ACTIVE VOCAB BANK, Replacement System)

REMOVE WORD INTERACTION

STATE: Default
- "Remove" button is visible for each word row

STATE: Remove Selected
- Tapped row becomes visually highlighted
- Show inline confirmation prompt:
  "Remove this word?"

- Show two inline actions:
  - "Confirm"
  - "Cancel"

STATE: Confirm Removal
- Word is removed from ACTIVE VOCAB BANK
- Replacement word is inserted immediately
- Show temporary inline message on that row position:
  "Replaced with: [new word]"

- Message fades after short delay
- Updated row remains in normal list state

STATE: Cancel Removal
- Row returns to normal state
- No changes made to bank

REPLACEMENT BEHAVIOR

When removal is confirmed:

1. Remove word from ACTIVE BANK
2. Select replacement word using SYSTEM — ACTIVE VOCAB BANK (V1)
3. Insert replacement into ACTIVE BANK as:
   - status = NEW
   - high frequencyWeight
4. Show temporary feedback:
   "Replaced with: [new word]"

List updates immediately

TAP "Add Word"

→ Go to SCREEN 0005 — ADD CUSTOM WORD

RETURN FROM ADD WORD

If word is successfully added:

- Word is added to MASTER DATASET
- Word is inserted into ACTIVE BANK
- One existing word is removed (replacement rule applies)

TAP "Done"

→ Return to previous screen

TRANSITIONS

Back > Return to previous screen
Home > Screen 0001
Add Word > Screen 0005
Done > Return to previous screen

STATE VARIABLES REQUIRED

- activeVocabList
- selectedLanguage

BEHAVIOR / LOGIC

- List must always reflect ACTIVE VOCAB BANK only
- Order may be:
  - random
  OR
  - grouped (optional future)

- Removing a word must NOT break:
  - POS distribution rules
  - competency compatibility

If removal would break rules:

→ system must select replacement that restores balance

CONSTRAINTS

- No editing of word fields in V0
- No filtering in V0
- No sorting in V0
- No frequency UI controls in V0

NOTES FOR IMPLEMENTATION

- Replacement selection must follow the V0 active-vocabulary-bank replacement rules
- V0 implementation must not depend on V1-only system documents

- This screen must work as a shared management screen for both V0 and later V1 flows
- Navigation should be launcher-based, not mode-select-based

DESIGN GOAL

Maximize:
- user control
- clarity
- simplicity

Minimize:
- friction
- complexity
- decision overload

MENTAL MODEL

"This is what I’m learning right now."

"I can swap things in and out easily."

END SCREEN

SCREEN 5101 — SPANISH VOCAB DECK (Not for V0)

TYPE
Management / List

LANGUAGE CONTEXT
Spanish

VISIBLE TEXT
"Your Vocabulary Deck"

LAYOUT

Top:
- Title: "Your Vocabulary Deck"

Middle:
- Vocab list

Bottom:
- Button: "Save"
- Button: "Back"

SAVE BUTTON STATE

Default: disabled (grayed out)
Enabled when at least one vocab item changes state (Green, Yellow, or Red)
Re-disabled if all changes are reverted to original state

BACK BUTTON BEHAVIOR

If no changes:
- Return immediately to previous screen

If unsaved changes exist:
- Prompt:
  "Discard changes?"
  - Button: "Discard"
  - Button: "Cancel"

SCROLL BEHAVIOR

Scrollable region: vocab list
Direction: vertical
Scroll indicator: visible

LIST ITEM — VOCAB ENTRY

Content:
- Target language word
- English translation

Control:
- Status button

STATUS BUTTON STATES

Green:
- Baseline state
- Word remains in deck at normal frequency

Yellow:
- Reduced-frequency state
- Word remains in deck, but appears less often

Red:
- Remove/replace state
- Word is marked for removal from the current deck

INTERACTION

Status button behavior:
- First tap: Green > Yellow
- Second tap: Yellow > Red
- Third tap: Red > Green

STATE RULES

Green:
- Word remains at normal frequency

Yellow:
- Word remains in deck
- Frequency is reduced in drills/study (via weighting system)

Red:
- Word is marked for replacement
- No replacement prompt appears during deck editing

SAVE BEHAVIOR

When the user taps "Save":
- Yellow states are committed
- Red-marked words are collected into a replacement queue

If no Red words:
- Changes are saved
- Return to previous screen

If one or more Red words:
- Enter replacement flow

POST-SAVE REPLACEMENT FLOW

- Red-marked words are processed one at a time
- Each word must be resolved before continuing

For each word:

"Select a replacement:"
- Button: "Choose from word bank"
- Button: "Add my own word"

REPLACEMENT FLOW RULES

- Replacement is required for all Red-marked words
- User cannot exit replacement flow with unresolved items
- Back navigation returns to replacement flow until all items are resolved

COMPLETION BEHAVIOR

After all replacements are completed:
- Changes are finalized
- Return to previous screen

TRANSITIONS

Choose from word bank > Screen Placeholder
Add my own word > Screen 000Z
Back > Screen Placeholder
Save > Screen Placeholder

SCREEN 5102 — FRENCH VOCAB DECK (Not for V0)

TYPE
Management / List

LANGUAGE CONTEXT
French

VISIBLE TEXT
"Your Vocabulary Deck"

LAYOUT

Top:
- Title: "Your Vocabulary Deck"

Middle:
- Vocab list

Bottom:
- Button: "Save"
- Button: "Back"

SAVE BUTTON STATE

Default: disabled (grayed out)
Enabled when at least one vocab item changes state (Green, Yellow, or Red)
Re-disabled if all changes are reverted to original state

BACK BUTTON BEHAVIOR

If no changes:
- Return immediately to previous screen

If unsaved changes exist:
- Prompt:
  "Discard changes?"
  - Button: "Discard"
  - Button: "Cancel"

SCROLL BEHAVIOR

Scrollable region: vocab list
Direction: vertical
Scroll indicator: visible

LIST ITEM — VOCAB ENTRY

Content:
- Target language word
- English translation

Control:
- Status button

STATUS BUTTON STATES

Green:
- Baseline state
- Word remains in deck at normal frequency

Yellow:
- Reduced-frequency state
- Word remains in deck, but appears less often

Red:
- Remove/replace state
- Word is marked for removal from the current deck

INTERACTION

Status button behavior:
- First tap: Green > Yellow
- Second tap: Yellow > Red
- Third tap: Red > Green

STATE RULES

Green:
- Word remains at normal frequency

Yellow:
- Word remains in deck
- Frequency is reduced in drills/study (via weighting system)

Red:
- Word is marked for replacement
- No replacement prompt appears during deck editing

SAVE BEHAVIOR

When the user taps "Save":
- Yellow states are committed
- Red-marked words are collected into a replacement queue

If no Red words:
- Changes are saved
- Return to previous screen

If one or more Red words:

- Enter replacement flow

POST-SAVE REPLACEMENT FLOW

- Red-marked words are processed one at a time
- Each word must be resolved before continuing

For each word:

"Select a replacement:"
- Button: "Choose from word bank"
- Button: "Add my own word"

REPLACEMENT FLOW RULES

- Replacement is required for all Red-marked words
- User cannot exit replacement flow with unresolved items
- Back navigation returns to replacement flow until all items are resolved

COMPLETION BEHAVIOR

After all replacements are completed:
- Changes are finalized
- Return to previous screen

TRANSITIONS

Choose from word bank > Screen 000Y
Add my own word > Screen 0005
Back > Screen 000X
Save > Screen 000X

SCREEN 5103 — GERMAN VOCAB DECK (Not for V0)

TYPE
Management / List

LANGUAGE CONTEXT
German

VISIBLE TEXT
"Your Vocabulary Deck"

LAYOUT

Top:
- Title: "Your Vocabulary Deck"

Middle:
- Vocab list

Bottom:
- Button: "Save"
- Button: "Back"

SAVE BUTTON STATE

Default: disabled (grayed out)
Enabled when at least one vocab item changes state (Green, Yellow, or Red)
Re-disabled if all changes are reverted to original state

BACK BUTTON BEHAVIOR

If no changes:
- Return immediately to previous screen

If unsaved changes exist:
- Prompt:
  "Discard changes?"
  - Button: "Discard"
  - Button: "Cancel"

SCROLL BEHAVIOR

Scrollable region: vocab list
Direction: vertical
Scroll indicator: visible

LIST ITEM — VOCAB ENTRY

Content:
- Target language word
- English translation

Control:
- Status button

STATUS BUTTON STATES

Green:
- Baseline state
- Word remains in deck at normal frequency

Yellow:
- Reduced-frequency state
- Word remains in deck, but appears less often

Red:
- Remove/replace state
- Word is marked for removal from the current deck

INTERACTION

Status button behavior:
- First tap: Green > Yellow
- Second tap: Yellow > Red
- Third tap: Red > Green

STATE RULES

Green:
- Word remains at normal frequency

Yellow:
- Word remains in deck
- Frequency is reduced in drills/study (via weighting system)

Red:
- Word is marked for replacement
- No replacement prompt appears during deck editing

SAVE BEHAVIOR

When the user taps "Save":
- Yellow states are committed
- Red-marked words are collected into a replacement queue

If no Red words:
- Changes are saved
- Return to previous screen

If one or more Red words:
- Enter replacement flow

POST-SAVE REPLACEMENT FLOW

- Red-marked words are processed one at a time
- Each word must be resolved before continuing

For each word:

"Select a replacement:"
- Button: "Choose from word bank"
- Button: "Add my own word"

REPLACEMENT FLOW RULES

- Replacement is required for all Red-marked words
- User cannot exit replacement flow with unresolved items
- Back navigation returns to replacement flow until all items are resolved

COMPLETION BEHAVIOR

After all replacements are completed:
- Changes are finalized
- Return to previous screen

TRANSITIONS

Choose from word bank > Screen Placeholder
Add my own word > Screen 0005
Back > Screen Placeholder
Save > Screen Placeholder

SCREEN XXXX — PLACEHOLDER

TYPE
Navigation / Competency Select

VISIBLE TEXT
“Test”

BUTTON GROUP: PRIMARY_ACTIONS
ID: Home
Label: "Home"

UI STYLE

PRIMARY_ACTIONS
Shape: rectangular with rounded corners
Label font: bold
Label size: medium
Position: centered, stacked
Spacing: medium

TRANSITIONS
Home > Screen 0001

GLOBAL SYSTEM — FULL PHRASE ANSWER VALIDATION

TYPE
Global Validation Rule

PURPOSE
Define how the app evaluates user-typed full-sentence or full-phrase answers on all "type full answer" cards.

SCOPE
Applies to:
- Full Phrase Production cards
- Any future card requiring typed full-answer production
- Any card that explicitly references this global validation system

DOES NOT APPLY TO
- Fill-in-the-blank cards
- Multiple-choice cards
- Tile-selection cards
- Any card with its own explicitly defined validation system

CORE PRINCIPLE
The app does not interpret language freely.
It checks whether the user produced one of the pre-approved correct target phrases.

VALIDATION INPUT
Each relevant card must provide:

1. CANONICAL ANSWER
- The primary expected German answer
- Example:
  "auf meinem Tisch"

2. ACCEPTED ALTERNATIVE ANSWERS (OPTIONAL)
- Pre-approved alternate German answers that are also correct for the English prompt
- These are lexical alternatives, not open-ended synonyms
- Example:
  Canonical: "Das ist wirklich gut."
  Accepted alternative: "Das ist echt gut."

3. NOUN TOKEN LIST
- A list identifying which word(s) in the answer are nouns
- Used to enforce capitalization rules

4. CONTRACTION-EQUIVALENT FORMS
- Stored internally so contracted and uncontracted forms can be matched as equivalent
- Example:
  "zu dem" ↔ "zum"
  "in dem" ↔ "im"
  "an dem" ↔ "am"
  "bei dem" ↔ "beim"
  "von dem" ↔ "vom"

5. SPECIAL-PROMPT FLAGS (OPTIONAL)
- Example:
  PROMPT_CONTAINS_YOUR = TRUE

NORMALIZATION RULES

Before validation, the app must:

- Trim leading and trailing spaces
- Collapse repeated internal spaces to a single space
- Ignore final punctuation only:
  - period
  - comma
  - exclamation point
  - question mark

DO NOT:
- Convert to lowercase
- Normalize umlauts
- Treat ä as ae
- Treat ö as oe
- Treat ü as ue
- Treat ß as ss

CAPITALIZATION RULES

GENERAL RULE
- German nouns must be capitalized correctly
- If a required noun is lowercase, the answer is incorrect

EXAMPLES
Correct:
- "mit meinem Hund"
- "Ich sehe den Tisch"

Incorrect:
- "mit meinem hund"
- "Ich sehe den tisch"

NON-NOUNS
- Non-noun capitalization is not required unless it begins the sentence
- Sentence-initial capitalization is allowed but is not by itself required unless that first word is also a noun

TOKEN MATCHING RULE

After normalization, the app must compare the user's response against:
- the canonical answer
- any accepted alternative answers
- contraction-equivalent versions of each approved answer

A response is correct only if:
- token count matches
- token order matches
- each token matches an approved answer path exactly
- noun capitalization is correct

NOT ALLOWED
- extra words

- missing words
- different word order
- partially correct phrases
- user-invented alternatives not explicitly approved for that card

CONTRACTION RULES

The app must accept contracted and uncontracted forms as equivalent when they are grammatically identical.

ACCEPT AS EQUIVALENT
- zu dem ↔ zum
- in dem ↔ im
- an dem ↔ am
- bei dem ↔ beim
- von dem ↔ vom

RULE
- If the user enters either the contracted or uncontracted approved form, mark it correct
- This applies only where the contraction is grammatically valid

DISPLAY RULE
- A card may display either the contracted or uncontracted form
- The validation engine must still accept the equivalent approved counterpart

LEXICAL ALTERNATIVE RULES

DEFINITION
Lexical alternatives are pre-approved alternate wordings that are genuinely correct translations of the English prompt.

EXAMPLE
English prompt:
- "That is really good."

Approved German answers:
- "Das ist wirklich gut."
- "Das ist echt gut."

RULES
- Lexical alternatives must be defined in advance at the card level
- The app must not generate new alternatives on its own
- If the user enters an approved lexical alternative, the answer is marked correct

POST-ANSWER DISPLAY FOR LEXICAL ALTERNATIVES
If the user enters an approved alternative that is not the canonical answer:
- Mark the answer correct
- Display:
  "Also accepted: [canonical answer]"

DO NOT DISPLAY
- "Also accepted: [user answer]"

SPECIAL RULE — ENGLISH "YOUR"

If the English prompt includes "your," the following possessive stems may all be accepted when grammatically correct for the sentence:

- dein-
- ihr-
- euer-

RULE
- All three are accepted if they produce a grammatically correct answer
- The rest of the phrase must still match an approved answer structure

POST-ANSWER DISPLAY
If the user enters one valid "your" form:
- Mark the answer correct
- Display a note at the bottom:
  "Also possible: [other valid form 1], [other valid form 2]"

VISUAL OUTPUT RULES

GENERAL PRINCIPLE
- Validation feedback must update existing feedback areas
- Validation must not create duplicate versions of the typed answer
- Validation must not create duplicate copies of the sentence elsewhere on the card unless the card spec explicitly says to do so

CORRECT RESPONSE
- Show:
  "Correct!"
- If user entered an approved lexical alternative instead of canonical answer:
  also show:
  "Also accepted: [canonical answer]"
- If user entered one valid "your" form where multiple are allowed:
  also show:

"Also possible: [other valid form 1], [other valid form 2]"

INCORRECT RESPONSE
- Show:
  "Incorrect — correct answer: [canonical answer]"
- If applicable, also show accepted alternatives beneath it:
  "Also accepted: [approved alternative 1], [approved alternative 2]"

LAYOUT RULE
- Feedback must appear in the card's fixed feedback area
- The card layout must not jump or reflow unnecessarily
- Existing sentence text should remain in place
- User input should remain visible unless the specific card spec says otherwise

ERROR DETECTION RULE

For incorrect answers, the app must identify the first token that fails to match an approved answer path.

ERROR TYPES TO DETECT
- Wrong article / possessive pronoun
- Wrong case form
- Wrong gender form
- Wrong noun capitalization
- Wrong lexical choice
- Wrong word order
- Extra token
- Missing token
- Wrong verb form
- Wrong noun form

PURPOSE OF ERROR DETECTION
- Supports internal analytics
- Supports future targeted feedback
- Supports card-specific feedback systems where applicable

V1 REQUIREMENT
- Error detection is required in v1
- Even if the user sees only final feedback, the system must still classify the error internally

MATCH ORDER

The app must validate in this order:

1. Compare against canonical answer
2. Compare against contraction-equivalent canonical answer
3. Compare against each accepted lexical alternative
4. Compare against contraction-equivalent versions of accepted alternatives
5. If prompt contains "your," compare against all valid approved possessive-variant forms
6. If no approved path matches, mark incorrect

CARD-LEVEL RESPONSIBILITY

Each "type full answer" card must explicitly define:
- canonical answer
- accepted lexical alternatives, if any
- whether "your" logic applies
- noun token list
- whether any accepted contraction equivalents apply

GLOBAL ENGINE RESPONSIBILITY

This global validation system is responsible for:
- normalization
- token comparison
- contraction equivalence
- capitalization enforcement
- lexical-alternative acceptance
- "your" variant acceptance
- error classification
- final validation result

OUTPUT STATES

PARTIAL CORRECTNESS RULE

Partial correctness is card-type dependent.

Allowed:
- matching cards
- multi-option recognition cards
- guided study cards where preserving correct subcomponents supports learning

Not allowed:
- strict drill constructed-response cards
- challenge cards
- assessment cards

In strict retrieval contexts, answers are evaluated as whole approved paths, not partially correct assemblies.

VALIDATION RESULT: CORRECT
- status = correct
- matched_form = canonical OR approved_alternative OR contraction_equivalent OR your_variant
- display primary feedback = "Correct!"
- display secondary feedback if applicable

VALIDATION RESULT: INCORRECT
- status = incorrect
- error_type = classified
- first_error_token_index = stored
- display primary feedback = "Incorrect — correct answer: [canonical answer]"
- display secondary accepted-answer text if applicable

NOTES FOR IMPLEMENTATION

- This is a rule engine, not free-language interpretation
- All accepted answers must be pre-authored
- No AI guessing
- No fuzzy synonym generation
- No umlaut simplification
- Noun capitalization is mandatory
- Contracted and uncontracted equivalents must be treated as matching only when explicitly valid

SYNONYM ACCEPTANCE RULE (GLOBAL)

- The system may ACCEPT user answers that use valid synonyms not present in the active vocabulary bank

- HOWEVER:
  - Card generation must ALWAYS use words from the active vocabulary bank
  - Synonyms must NEVER be introduced by the system as:
    - prompts
    - correct answers
    - distractors

- If a user enters a valid synonym:
  - Mark answer as correct
  - Display:
    "Also accepted: [canonical answer from vocab bank]"

- Synonyms do NOT:
  - enter the repetition system
  - replace canonical vocabulary items
  - affect spaced repetition or weighting

DESIGN INTENT

- Preserve user correctness
- Protect repetition integrity
- Prevent vocabulary drift

ROMANCE-LANGUAGE STUDY MODE ARTICLE + COLOR RULE

French and Spanish Study Mode nouns must support article + color pairing.

RULES

- On noun-based Study cards in French and Spanish, the article is displayed with the noun
- The noun itself is color-coded by gender
- The article is not color-coded
- Displayed article may be definite or indefinite depending on the card’s generation rules
- Canonical storage remains bare lemma only

PURPOSE

- The visible article reinforces the noun’s gender
- The noun color reinforces the same pattern visually
- Article support and color support work together, not as alternatives

GLOBAL UI RULES

Language buttons always display the national flag of the language's primary country.

Language button flag behavior:

Flags are flush with the left edge of the button and occupy the full height of the button.
Flags are not padded.
The top-left and bottom-left corners of the flag are clipped by the button’s rounded corners.
Flag aspect ratio is preserved.

Status button color meanings:

Green = normal frequency
Yellow = reduced frequency
Red = marked for replacement

GERMAN STUDY MODE COLOR RULE

- In German Study Mode, nouns are color-coded by default on exposure cards

- Color applies only to the noun itself, never to the full phrase or sentence

- German vocab-only exposure cards must not display der, die, or das to indicate gender

- In German Drill Mode, nouns are not color-coded by default when gender knowledge is being tested

- In lower-pressure Drill cards, color may appear only as corrective feedback when the user's error is specifically gender-related

- In higher-pressure Drill cards and assessments, no noun color support is shown

GERMAN RECOGNITION RULE

- Articles and pronouns are tested as output forms only
- Gender is NOT learned through articles
- Gender encoding remains color-based only

German nouns must always appear capitalized.

Gender color system:
masculine = #4da3ff
feminine = #d66aa6
neuter = #A4922D